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ENGINEERING



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Saveetha Institute of Medical And Technical Sciences
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Identity and Access Management (IAM) in Cloud Environments

Course Code: CSA1507

Course Name: Cloud Computing and Big Data analytics

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Introduction

- Identity and Access Management (IAM) is a security framework used to manage user identities and control access to cloud resources.
 - It ensures that only authorized users, devices, and applications can access cloud services and data.
 - IAM verifies the identity of users through authentication and determines their access rights through authorization.
 - It protects cloud environments from unauthorized access, data breaches, and cyber threats.
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- IAM follows the principle of "the right user gets the right access to the right resource at the right time."



Problem Statement

- Cloud environments are accessed by multiple users, devices, and applications, increasing security risks.
- Traditional username and password authentication is not sufficient to protect cloud resources.
- Unauthorized access can lead to data breaches, identity theft, and financial losses.
- Managing user identities and permissions becomes complex as organizations grow.
- Incorrect or excessive user permissions can expose sensitive business data.



Objective

- To develop a real-time kernel monitoring and visualization system.
- To capture and trace system calls generated by running processes.
- To visualize process interactions and kernel activities through an interactive interface.
- To map system resources such as CPU, memory, files, and network connections.
- To provide a clear understanding of how applications interact with the operating system kernel.



Core Components of IAM

Identity:

- Users
- Devices
- Applications

Authentication:

- Password
- OTP
- Biometrics
- Multi-Factor Authentication (MFA)

Authorization:

- Grants or denies access.

Policies:

- Define permissions.

Roles:

- Admin
- Developer
- Viewer

Auditing:

- Records user activities

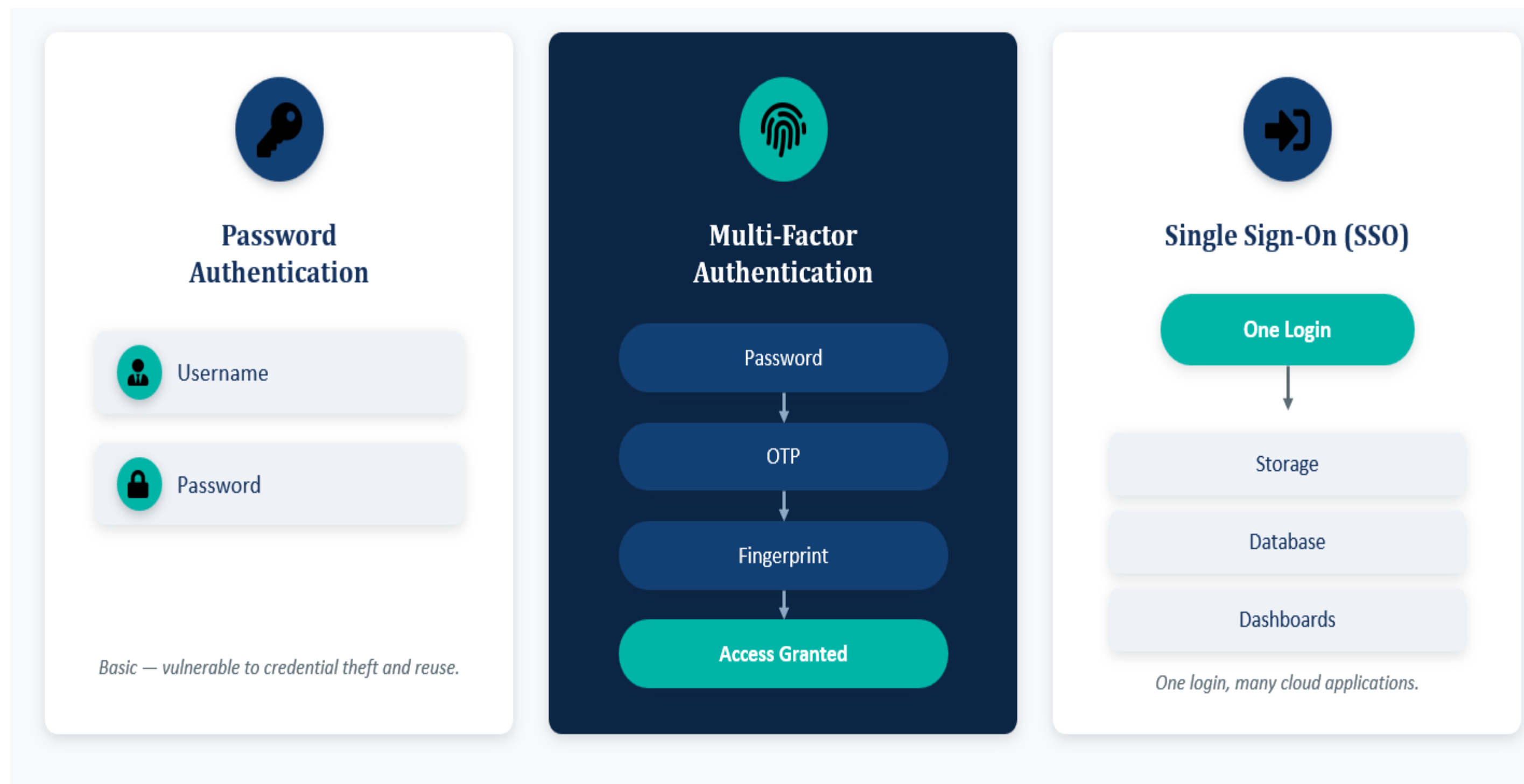


IAM Architecture





Authentication method





Authorization Access and Control

Role-Based Access Control (RBAC)

Role	Permission
Student	Read
Faculty	Read & Write
Developer	Deploy
Admin	Full Access

Access is determined by the role assigned to a user — simple, predictable, and easy to audit.

Attribute-Based Access Control (ABAC)

Access depends on multiple attributes evaluated together:



Department



Location



Time



Device

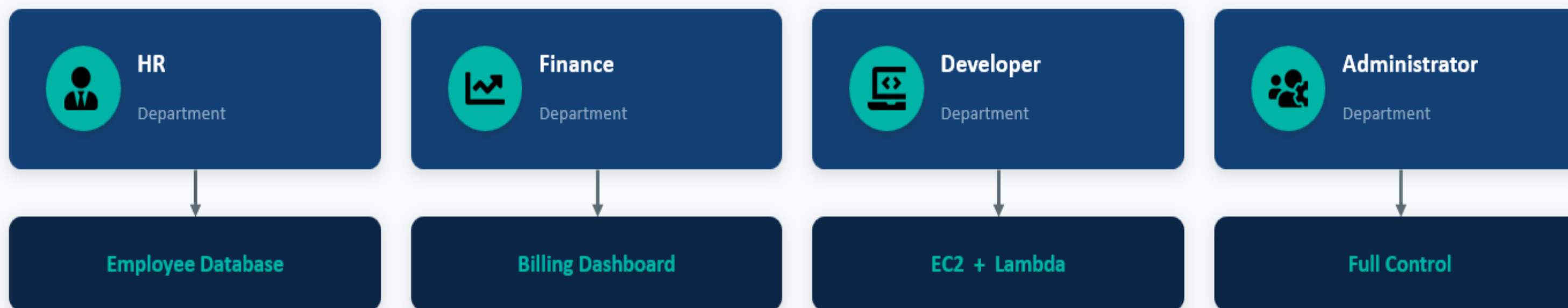
Example: Department = HR AND Office Network

Access Granted



Real World Example

How company departments map to controlled cloud resource access.



SECURITY FEATURES

- ✓ IAM Roles & Policies
- ✓ Multi-Factor Authentication
- ✓ CloudTrail Logging

RESULT

- 🛡️ Secure cloud infrastructure
- 🛡️ Controlled, least-privilege access
- 🛡️ Regulatory compliance



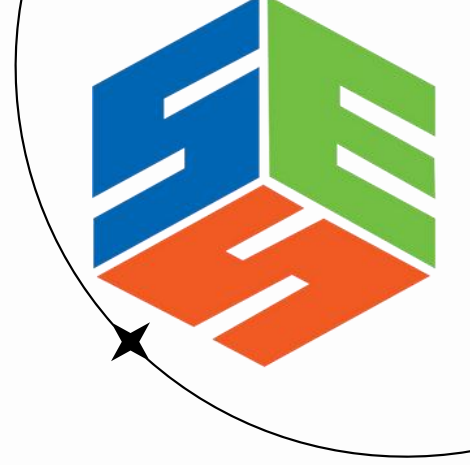
Conclusion

- IAM is the foundation of cloud security. It authenticates users and authorizes access to cloud resources.
- Technologies like MFA, RBAC, ABAC, and SSO improve cloud security.
- IAM protects sensitive data and supports regulatory compliance.
- Every modern cloud platform relies on IAM for secure and efficient access management



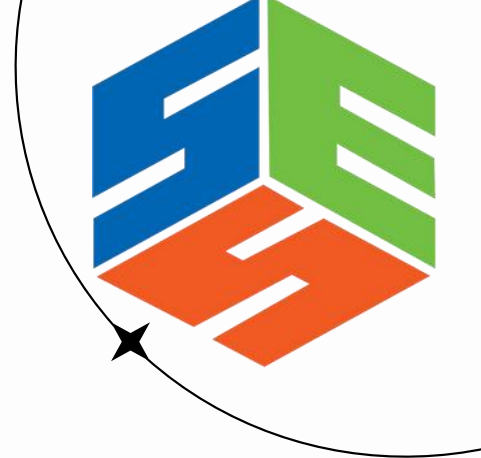
Future Scope

- Integration of Artificial Intelligence and Machine Learning for advanced anomaly detection and predictive performance analysis.
- Support for cloud, containerized, and distributed systems to monitor large-scale computing environments.
- Implementation of automated troubleshooting and self-healing mechanisms based on detected kernel events.
- Enhancement of visualization features with 3D graphs and advanced analytics dashboards for deeper system insights.

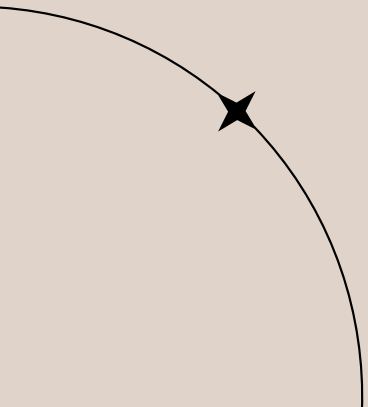


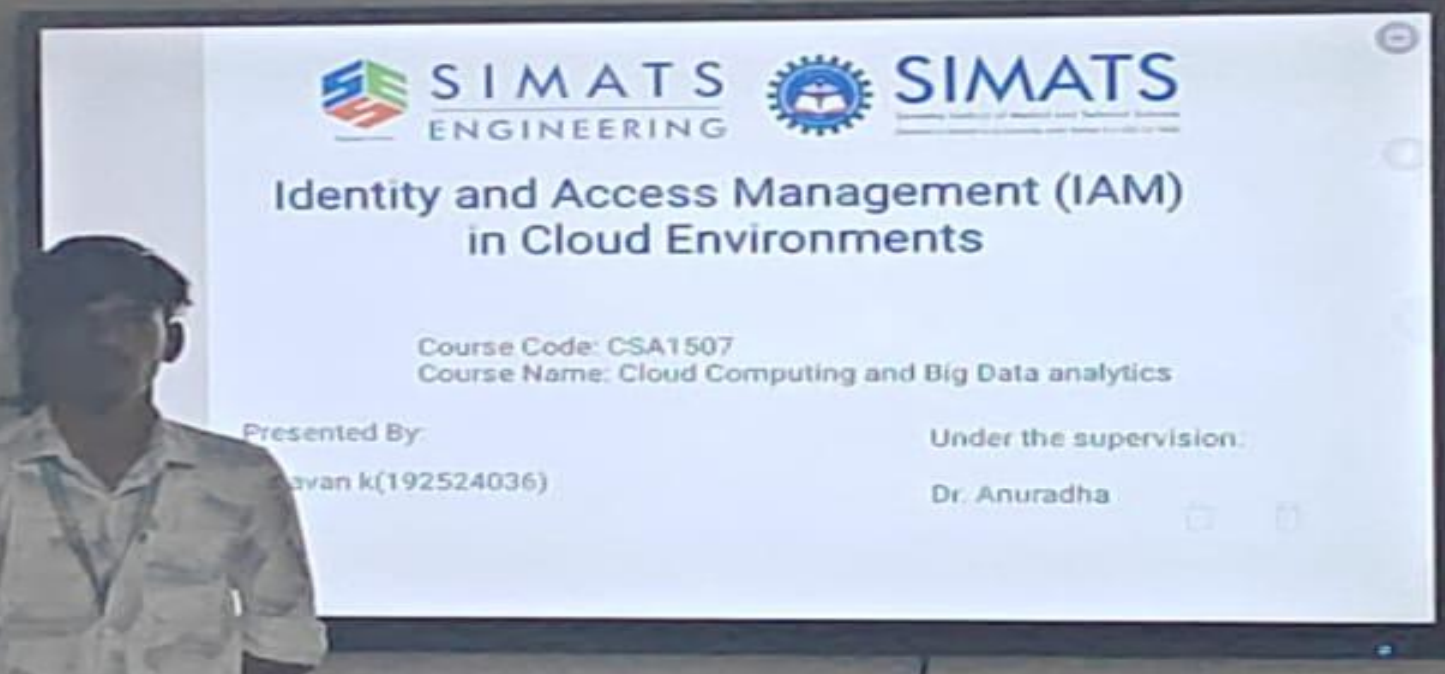
Reference

- T. Erl, R. Puttini, and Z. Mahmood, Cloud Computing: Concepts, Technology & Architecture. Pearson, 2014.
- R. Buyya, C. Vecchiola, and S. T. Selvi, Mastering Cloud Computing. Morgan Kaufmann, 2013.
- Amazon Web Services, AWS Identity and Access Management (IAM).
Microsoft, Microsoft Entra ID Documentation.
- Google Cloud, Cloud Identity and Access Management (IAM).



THANK YOU ✨





GPS Map Camera



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